MIL-E-1/1073F
19 February 1981
SUPERSEDING
MIL-E-1/1073E
31 January 1975

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, RECEIVING

TYPE 5744WB

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Triode, subminiature, high Mu

Outline	---	3-6 (EIA)
Base	---	Pinch press (5 leads in line)
Envelope	---	T3
Cathode	---	Coated unipotential

Base connections:

Pin No.	---	1	2	3	4	5
Element	---	a (colored dot)	h	h	g	k

ABSOLUTE-MAXIMUM RATINGS:

Parameter:	Ef	Eb	Ec	Ehk	Rk	Rg	Ib	Ic	Pp	TE	Alt
Unit:	V	Vdc	Vdc	v	Ohms	Meg	mAdc	mAdc	W	C	ft
Maximum:	6.9	275	---	200	---	1.2	6.5	1.0	1.4	220	(note 1)
Minimum:	5.7	---	-55	---	---	---	0.5	---	---	---	---
TEST CONDITIONS:	6.3	250	0	0	500	---	---	---	---	---	---

GENERAL:

Qualification - Required

Reliable tube

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits		Unit
							Min	Max	
	Quality conformance inspection, part 1								
1256	Electrode current (1) (anode)			0.4	II	Ib	3.2	5.2	mAdc
1256	Electrode current (2) (anode)		Ec = -6.5 Vdc	0.4	II	Ib	---	50	µAdc
1266	Total grid current	3		0.4	II	Ic	---	-0.3	µAdc
1301	Heater current			0.4	II	If	190	210	mA
1306	Transconductance (1)			0.4	II	Sm	3,400	4,600	µmhos
1321	AC amplification		Esig = 0.2 Vac; Ebb = 100 Vdc; Ecc = 0; Rk = 0	0.4	II	Ep	6.5	---	Vac
1336	Heater-cathode leakage			0.4	II	Ihk	---	5	µAdc
1201	Short and discontinuity detection			0.4	II	---	---	---	---
	Quality conformance inspection, part 2								
1211	Insulation of electrodes			2.5	S3	R	---	---	---
1031	High-frequency vibration		Rp = 10,000 ohms; Ck = 1,000 µF	2.5	I	Ep	---	15	mVac
1031	Sweep-frequency vibration		Rp = 10,000 ohms; Ck = 1,000 µF; F = 30 to 1,000 Hz; 15 G; t = 3 minutes	4.0	S3	ep	---	75	mv p to p
1246	Audio frequency noise	4	Esig = 50 mVac; Rp = 0.2 Meg; Rg = 1.0 Meg;	2.5	I	EB	---	---	---
1256	Electrode current (3) (anode)		Ec = -4.0 Vdc	2.5	I	Ib	5	---	µAdc
1266	Grid emission	5	Ef = 7.5 V; Ec = -10 Vdc	---	---	Is	---	-0.4	µAdc
1306	Transconductance (2)		Ef = 5.7 V	2.5	I	ΔSm Ef	---	5	%
1316	Amplification factor			---	---	Mu	60	80	---
1331	Direct-interelectrode capacitance		0.405 inch dia. shield	6.5	Code E	Cgp Cin Cout	0.65 2.0 1.6	0.95 3.4 3.0	pF pF pF

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Method	Requirement or test	Notes	Conditions	AQL (percent defective)	Inspection level or code	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 2 - Continued								
1256	Electrode current (1) (anode) (tube operation time)		t = 20 seconds	4.0	S3	Ib	90	110	%
1116	Lead fatigue			2.5	Code F	---	---	---	---
2126	Envelope strain			2.5	I	---	---	---	---
1041	Shock	6	400 G; Ehk = 100 Vdc	---	---	---	---	---	---
1031	Vibration-fatigue (1)			6.5	See note 7	---	---	---	---
1031	Vibration-fatigue (2)		10 G; t = 6 hours	6.5	See note 7	---	---	---	---
---	Shock and vibration-fatigue (1) and (2)-test end points:								
1031	High-frequency vibration			---	---	Ep	---	25	mVac
1336	Heater-cathode leakage			---	---	Ihk	---	10	µAdc
1306	Change in transconductance (1) of individual tubes			---	---	ΔSm t	---	10	%
1266	Total grid current			---	---	Ic	---	-1.0	µAdc
1105	Permanence of marking			---	---	---	---	---	---

Symbol	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3						
1506	Heater-cycling life		Ef = 7.5 V; Ehk = 140 Vac; Ec = 0; Eb = 0	---	---	---	---
---	Heater-cycling life-test end point:						
1336	Heater-cathode leakage			Ihk	---	20	µAdc
1516	Stability life		Ehk = 200 Vdc; Rgl = 1.0 Meg; TA = room	---	---	---	---
---	Stability life-test end point:						
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	10	%

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Method	Requirement or test	Notes	Conditions	Symbol	Limits Min	Limits Max	Unit
	Quality conformance inspection, part 3 - Continued						
1501	Intermittent life	2	Ehk = 200 Vdc; Rg1 = 1.0 Meg; TA = room; TE = 220°C (min); Group E	---	---	---	---
---	Intermittent life-test end points (500 hours):	8					
1266	Total grid current			Ic	0	-0.6	µAdc
1301	Heater current			If	183	217	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	20	%
1306	Transconductance (2)			ΔSm Ef	---	15	%
1336	Heater-cathode leakage			Ihk	---	10	µAdc
1211	Insulation of electrodes			R	25	---	Meg
1306	Transconductance (1), average change			Avg ΔSm t	---	15	%
---	Intermittent life-test end points (1,000 hours):	8					
1266	Total grid current			Ic	0	-0.6	µAdc
1301	Heater current			If	183	217	mA
1306	Change in transconductance (1) of individual tubes			ΔSm t	---	25	%
1306	Transconductance (2)			ΔSm Ef	---	20	%
1336	Heater-cathode leakage			Ihk	---	10	µAdc
1211	Insulation of electrodes			R	25	---	Meg

NOTES:

1. See "Reduced pressure (altitude) rating", and altitude, maximum peak voltage in the basic document.
2. Envelope temperature (TE) requirements, when measured in accordance with the temperature by conduction-band measurement (method 1226), will be satisfied if a TUT having bogey Ib (±5 percent) under normal test conditions, is determined to operate at or above minimum specified temperature at any position in the life-test rack.
3. This test shall be performed at the conclusion of the holding period.
4. The rejection level shall be set at the VU meter reading obtained during calibration.
5. Prior to this test, tubes shall be preheated a minimum of 5 minutes at the conditions indicated below. The 3-minute test shall not be permitted. Test at specified conditions, except the value of Ec shall be -10 Vdc, within 3 seconds after preheating. Grid emission shall be the last test performed on the sample selected for the grid emission test.

Ef V	Ec Vdc	Eb Vdc	Rk Ohms	Rg Meg
7.5	0	250	500	1.0

6. A grid resistor of 0.1 megohm shall be added; however, this resistor shall not be used when a thyratron-type short indicator is used.
7. This test shall be conducted on the initial lot and thereafter on a lot approximately every 12 months. When one lot has passed, the 12-month rule shall apply. In the event of lot failure, the lot shall be rejected and the succeeding lots shall be subjected to this test until a lot has passed. MIL-STD-105, sample size code letter E, shall apply.

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8. The allowable defectives per characteristic shall not be greater than 1 for the first sample and shall not be greater than 3 for combined samples. Total defectives shall be 3 and 6, respectively, for 500 hours and 4 and 8, respectively, for 1,000 hours.
9. Revision letters are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

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Air Force - 85

Review activities:
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AIR FORCE - 99
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User activities:
Navy - AS, OS, CG
Air Force - 11

Preparing activity:
Navy - EC

Agent:
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